

Notes: Adams, Einav, and Levin (AER, 2009)
Liquidity Constraints and Imperfect Information in Subprime Lending

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1 Big picture

- **Question.** The paper asks why subprime borrowers appear liquidity constrained and whether these constraints can be connected to the actual information problems faced by lenders.
- **Setting.** The empirical setting is a large used-car sales and finance company serving low-income, poor-credit borrowers in the United States from June 2001 to December 2004, with loan performance observed through April 2006.
- **Demand-side fact 1: tax-season liquidity.** Applications and sales rise sharply during tax rebate season. The increase is strongly related to the size of the Earned Income Tax Credit (EITC), especially for low-income households with dependents.
- **Demand-side fact 2: down-payment sensitivity.** A \$100 increase in the minimum down payment reduces purchase probability by roughly 3 percentage points, or about 9 percent of demand. To get the same demand reduction through the car price requires a price increase of almost \$3,000.
- **Supply-side fact 1: default is very high.** More than half of loans end in default. Default is often early, so the present value of loan payments is frequently below the original loan balance.
- **Supply-side fact 2: loan size affects default.** For a given borrower, a larger loan increases default. The paper estimates that a \$1,000 larger loan increases the default hazard by about 16 percent.
- **Supply-side fact 3: adverse selection.** Riskier borrowers want larger loans and smaller down payments. This selection occurs both across observable risk categories and within risk categories through unobservable borrower information.
- **Lender response.** The lender uses credit scoring, risk-based pricing, minimum down payments, interest-rate variation, and loan caps. These practices mitigate adverse selection but do not eliminate the liquidity constraints faced by borrowers.
- **Economic takeaway.** The paper connects two literatures: household liquidity constraints and asymmetric-information models of credit markets. Liquidity-constrained consumer behavior is not just a reduced-form demand fact; it is linked to lender-side moral hazard, adverse selection, and legal or institutional constraints on pricing.

2 Incremental contribution of the paper

- **From documenting liquidity constraints to explaining them.** Prior consumption studies show that low-wealth households respond to predictable cash flows. This paper adds direct evidence on the credit-market frictions that can generate those constraints.
- **Unique matched demand-and-repayment data.** The paper observes applications, purchases, prices, down payments, loan terms, credit categories, defaults, and recoveries from the same subprime lender. This allows demand behavior and repayment risk to be studied jointly.
- **Direct evidence on short-run cash sensitivity.** The tax-season evidence is unusually clean because the timing and size of EITC-related refunds create predictable temporary cash-on-hand shocks.

- **A sharp down-payment versus price comparison.** The paper compares sensitivity to an immediate payment with sensitivity to a deferred payment. The enormous gap between the two sensitivities is difficult to rationalize with ordinary impatience alone.
- **Empirical separation of moral hazard and adverse selection.** Both forces create a positive relation between loan size and default. The paper uses exogenous variation in car price and minimum down payment to isolate the causal effect of loan size on default, then uses residual loan-demand variation to measure self-selection.
- **Credit-market theory disciplined by data.** The simple model explains why lenders use loan caps, minimum down payments, risk-based pricing, and lower interest rates for borrowers who signal lower risk by paying more down.
- **Evidence on credit scoring.** Risk-based pricing reduces adverse selection across observable risk groups: high-risk borrowers would like larger loans, but the lender forces them to make larger down payments.
- **Practical policy contribution.** The paper shows why expanding credit access is not only a matter of lowering interest rates. Moral hazard and adverse selection make unconstrained lending unprofitable even when demand is strong.

3 Data and measurement

3.1 Subprime auto lending environment

- The data come from a used-car sales company that also originates loans for its customers.
- Customers are mostly low-income, subprime borrowers who generally cannot obtain conventional bank credit.
- The data include more than 50,000 applications; the exact number is suppressed for confidentiality.
- Each applicant is assigned a proprietary credit category. The notes below use the paper's broad low-, medium-, and high-risk categories.
- A typical loan has a high annual percentage rate, often close to state interest-rate caps, and a term of about 40 months.

Interpretation: The institutional environment is useful because it is exactly the market where liquidity constraints should be most visible. At the same time, the company observes much more information than is usually available to researchers.

3.2 Applications, purchases, and liquidity proxies

- The paper observes all applications, not only completed sales. This is critical for estimating purchase demand.
- Short-run liquidity is measured indirectly through tax rebate timing and through sensitivity to required down payments.
- The tax-season evidence uses heterogeneity in expected EITC rebates by income and dependents.

- Down-payment sensitivity uses variation in required minimum down payments across time, cities, and credit categories.

Interpretation: The paper does not need to observe household balance sheets directly. Instead, it infers liquidity constraints from behavior that should not arise for consumers who can freely borrow against future income.

3.3 Loan terms, default, and recoveries

- The paper observes transaction prices, down payments, interest rates, loan terms, monthly payments, defaults, and recovery amounts.
- More than half of loans with uncensored repayment histories default.
- Default tends to occur early, which means many borrowers make only a small fraction of scheduled payments.
- The company can lose money on marginal increases in loan size because larger loans increase default risk.

Interpretation: The high default rate is not a side detail. It is the reason the lender has a strong incentive to restrict loan size and require cash up front.

3.4 Credit categories and risk-based pricing

- The company uses proprietary credit scoring to sort borrowers into risk categories.
- Riskier borrowers face higher minimum down payments, which mechanically limits the loan size they can choose.
- Buyers who voluntarily pay more down can sometimes receive better interest rates.
- Credit scoring therefore plays two roles: it predicts default and it determines the contract menu offered to the borrower.

Interpretation: Credit scoring mitigates adverse selection across observable types. But the paper also finds adverse selection within risk categories, so scoring does not fully solve asymmetric information.

4 Models and empirical specifications

4.1 Purchase demand model

The purchase decision is modeled as a binary choice:

$$q_i = 1 \iff q_i^* = x_i' \alpha + \varepsilon_i > 0.$$

- The dependent variable is whether an applicant buys a car.
- Offer variables include negotiated price, minimum down payment, interest rate, and loan term.
- Car controls include cost, age, mileage, and time on lot.

- Applicant controls include income, age, bank account status, home ownership, living with parents, city, month, year, and credit category.
- Because nonbuyers do not have observed chosen cars and negotiated prices, the paper imputes cars for nonbuyers and also checks cell-level specifications that avoid explicit imputation.

Interpretation: The key empirical contrast is between the coefficient on the minimum down payment and the coefficient on car price. The former is paid immediately; the latter can be financed.

4.2 Price endogeneity and list-price instruments

Negotiated price may contain salesperson information or bargaining effects. The paper therefore estimates a negotiated-price equation:

$$p_i = \lambda l_i + x_i' \alpha_x + v_i,$$

where l_i is the centrally set list price.

- List price is driven by centrally set markup rules, not by the individual salesperson's negotiation with a specific applicant.
- The instrumental-variable approach uses variation in list prices, cost buckets, state-level interest-rate-cap rules, and markup-schedule changes.
- The paper also runs robustness checks around policy changes and discontinuities in the margin formula.

Interpretation: The demand evidence is not intended to be a simple correlation between price and sales. It uses institutional variation in company pricing and down-payment policies.

4.3 Down-payment and loan-demand model

For buyers, the paper models the desired down payment with a Tobit specification:

$$D_i = \begin{cases} D_i^* = x_i' \gamma + e_i, & \text{if } D_i^* \geq d_i, \\ d_i, & \text{if } D_i^* < d_i, \end{cases}$$

where d_i is the required minimum down payment.

- This is equivalently a model of loan demand: a lower down payment means a larger loan.
- A large fraction of buyers choose exactly the minimum, which motivates the censored Tobit model.
- The paper asks whether high-risk borrowers prefer smaller down payments and larger loans.

Interpretation: The down-payment model measures adverse selection in desired borrowing. If high-risk borrowers want larger loans, a common contract menu would attract worse risks to larger loan sizes.

4.4 Default hazard model and control function

The default model is a Cox proportional hazard model:

$$h(t \mid L_i, x_i) = \exp(L_i \delta_L + x_i' \delta_x) h_0(t),$$

where t is the fraction of scheduled loan payments made.

- The loan-size coefficient captures how the default hazard changes with the amount financed.
- To isolate moral hazard, the paper controls for the residual from the down-payment model.
- The down-payment residual proxies for private borrower information revealed by the borrower's chosen down payment.
- With this residual included, remaining loan-size variation is driven by price and minimum-down-payment policies rather than by borrower selection.

Interpretation: This is the central empirical innovation. It separates the causal effect of loan size on default from the fact that riskier people may choose larger loans.

4.5 Credit-market theory

The model has two periods. Borrowers choose a car loan in period 1 and decide whether to repay in period 2. A contract can be summarized by a down payment D and future payment M .

- Moral hazard means the probability of repayment falls as loan liability rises.
- Because larger loans increase default costs, firm profits can first rise and then fall with loan size.
- This creates a rationale for minimum down payments or loan caps, even under competition.
- Adverse selection means high-risk borrowers prefer lower down payments and larger loans.
- Risk-based pricing can separate borrowers and reduce cross-type selection.

Interpretation: The model explains why liquidity constraints can persist in equilibrium. The lender is not simply refusing profitable business; it is responding to default incentives and selection.

5 Identification and empirical design

5.1 Identifying liquidity constraints

- **Tax-season strategy.** The timing of tax rebates generates predictable liquidity variation. If borrowers could freely borrow against future income, demand should not jump sharply when the refund arrives.
- **EITC heterogeneity.** The tax-season demand spike is larger for groups with larger expected EITC rebates.
- **Down-payment versus price strategy.** A down payment is due immediately, while a higher price can be rolled into the loan. A much larger response to the immediate payment indicates liquidity constraints.

Interpretation: The demand evidence is strongest because it combines time-series seasonality, cross-group EITC variation, and policy variation in minimum down payments.

5.2 Identifying moral hazard

- Moral hazard is identified from variation in loan size that is not chosen by borrowers based on their private default risk.
- Exogenous variation comes from car prices, centrally set markups, and minimum down-payment rules.
- Conditional on borrower controls and the down-payment residual, a larger loan raises the default hazard.

Interpretation: The estimate is not just that larger loans default more. It is that increasing a given borrower's loan size makes default more likely.

5.3 Identifying adverse selection

- Observable adverse selection is measured by the fact that worse credit categories prefer lower down payments and larger loans.
- Unobservable adverse selection is measured by the correlation between the down-payment residual and default, conditional on loan size and controls.
- When the residual is omitted from the default hazard model, the loan-size coefficient rises, showing that selection contributes to the loan-size/default correlation.

Interpretation: The paper shows both ex ante selection across known risk groups and selection on private information within risk groups.

5.4 Identifying lender response

- The lender uses higher minimum down payments for higher-risk borrowers.
- Higher-risk buyers would like smaller down payments, but risk-based pricing pushes them toward larger down payments.
- The paper compares desired down payments with actual constrained down payments to show how risk-based pricing changes the correlation between default risk and loan size.

Interpretation: Credit scoring can reverse the raw selection pattern across risk groups. Without pricing rules, high-risk borrowers would take larger loans; with risk-based down-payment requirements, they are constrained to borrow less.

5.5 Limitations of the empirical design

- The data come from one company, so external validity to all subprime lending markets is not automatic.

- The paper observes rich application and loan data but not full household balance sheets, outside credit access, or borrowers' subjective expectations.
- The causal interpretation of moral hazard relies on the exclusion restriction that price and minimum-down-payment policies affect default only through loan size, conditional on controls.
- The paper studies equilibrium behavior under an existing institutional regime, including interest-rate caps and proprietary credit scoring.

Interpretation: The design is strongest for showing that liquidity constraints, moral hazard, and adverse selection are jointly important in this subprime auto-loan environment. It is less suited to distinguishing among deeper behavioral models of borrower default.

6 Tables and figures

6.1 Table 1: Summary statistics

Read:

- Table 1 summarizes applicants, buyers, cars, transactions, and loan outcomes.
- Applicants are low-income: the mean monthly income is roughly \$2,414.
- Only about one-third of applicants buy a car.
- The average car price is roughly \$10,777, while the average down payment is below \$1,000 and the average loan amount is roughly \$10,740.
- Default is high: among uncensored loans, the default rate is 61 percent.

Economic message: The table establishes the setting: borrowers have limited resources, loans are large relative to down payments, interest rates are high, and default is common. This is exactly the environment where liquidity constraints and lender-side information problems should matter.

TABLE 1—SUMMARY STATISTICS

	Number of observations ^a	Mean	Standard deviation	Fifth percentile	Ninety-fifth percentile
<i>Applicant characteristics</i>					
Age	<i>N</i>	32.8	10.7	19	53
Monthly income	<i>N</i>	2,414	1,074	1,299	4,500
House owner	<i>N</i>	0.15	—	—	—
Lives with parents	<i>N</i>	0.18	—	—	—
Bank account	<i>N</i>	0.72	—	—	—
Risk category					
Low	<i>N</i>	0.27	—	—	—
Medium	<i>N</i>	0.45	—	—	—
High	<i>N</i>	0.29	—	—	—
Car purchased	<i>N</i>	0.34	—	—	—
<i>Buyer characteristics</i>					
Age	0.34 <i>N</i>	34.7	10.8	20	55
Monthly income	0.34 <i>N</i>	2,557	1,089	1,385	4,677
House owner	0.34 <i>N</i>	0.17	—	—	—
Lives with parents	0.34 <i>N</i>	0.16	—	—	—
Bank account	0.34 <i>N</i>	0.76	—	—	—
Risk category					
Low	0.34 <i>N</i>	0.35	—	—	—
Medium	0.34 <i>N</i>	0.47	—	—	—
High	0.34 <i>N</i>	0.17	—	—	—
<i>Car characteristics</i>					
Acquisition cost	0.34 <i>N</i>	5,213	1,358	3,205	7,240
Total cost	0.34 <i>N</i>	6,096	1,372	4,096	8,213
Car age (years)	0.34 <i>N</i>	4.3	1.9	2	8
Odometer	0.34 <i>N</i>	68,776	22,091	31,184	102,300
Lot age (days)	0.34 <i>N</i>	33	44	1	122
Car price	0.34 <i>N</i>	10,777	1,797	8,095	13,595
<i>Transaction characteristics</i>					
Minimum down payment (applicants)	<i>N</i>	750	335	400	1,400
Minimum down payment (buyers)	0.34 <i>N</i>	648	276	400	1,200
Interest rate (APR)	0.34 <i>N</i>	26.2	4.4	17.7	29.9
Loan term (months)	0.34 <i>N</i>	40.5	3.7	35	45
Down payment	0.34 <i>N</i>	963	602	400	2,000
Loan amount	0.34 <i>N</i>	10,740	1,802	7,982	13,560
Monthly payment	0.34 <i>N</i>	395	49	314	471
Default (uncensored observations only)	0.13 <i>N</i>	0.61	—	—	—
Recovery amount (uncensored defaults)	0.08 <i>N</i>	1,382	1,386	0	3,784

^a To preserve the confidentiality of the company that provided the data, we do not report the exact number of observations, $N \gg 50,000$.

Table 1: Summary statistics.

6.2 Figures 1A and 1B: Default timing and return distribution

Read:

- Figure 1A shows that conditional on default, many borrowers default after paying only a small fraction of the loan.
- Figure 1B shows a bimodal distribution of returns: paid loans often produce positive returns, while defaulted loans generate low or negative returns.
- The distribution highlights why marginal loan dollars are risky for the lender.

Economic message: Default is not only frequent; it is often early. This makes loan size a central contracting variable because larger loans expose the lender to greater losses before much repayment occurs.

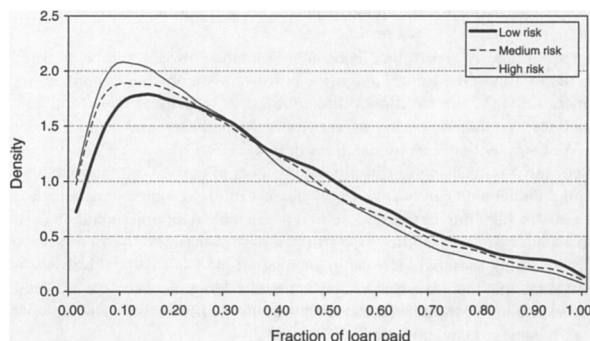


FIGURE 1A. KERNEL DENSITY OF FRACTION OF LOAN PAID CONDITIONAL ON DEFAULT

(a) Figure 1A: Fraction of loan paid conditional on default.

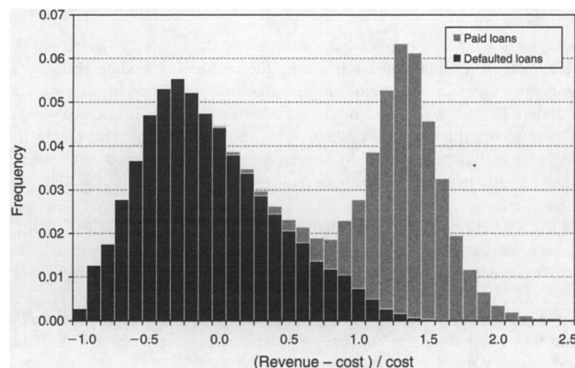


FIGURE 1B. RATE OF RETURN HISTOGRAM

(b) Figure 1B: Rate-of-return histogram.

6.3 Figures 2A and 2B: Tax-season demand and EITC heterogeneity

Read:

- Figure 2A shows a clear spike in applications and sales in late January and February.
- Figure 2B shows that the February increase is largest for income/dependent groups with the largest expected EITC refunds.
- Applications, sales, and close rates all rise more strongly for groups likely to receive larger rebates.

Economic message: The seasonal pattern is hard to explain by car needs alone. It is consistent with borrowers making purchases when cash arrives, which is a signature of liquidity constraints.

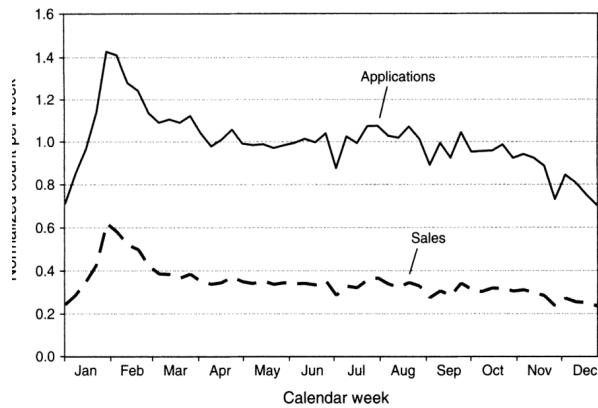


FIGURE 2A. SEASONALITY IN APPLICATIONS AND SALES

(a) Figure 2A: Seasonality in applications and sales.

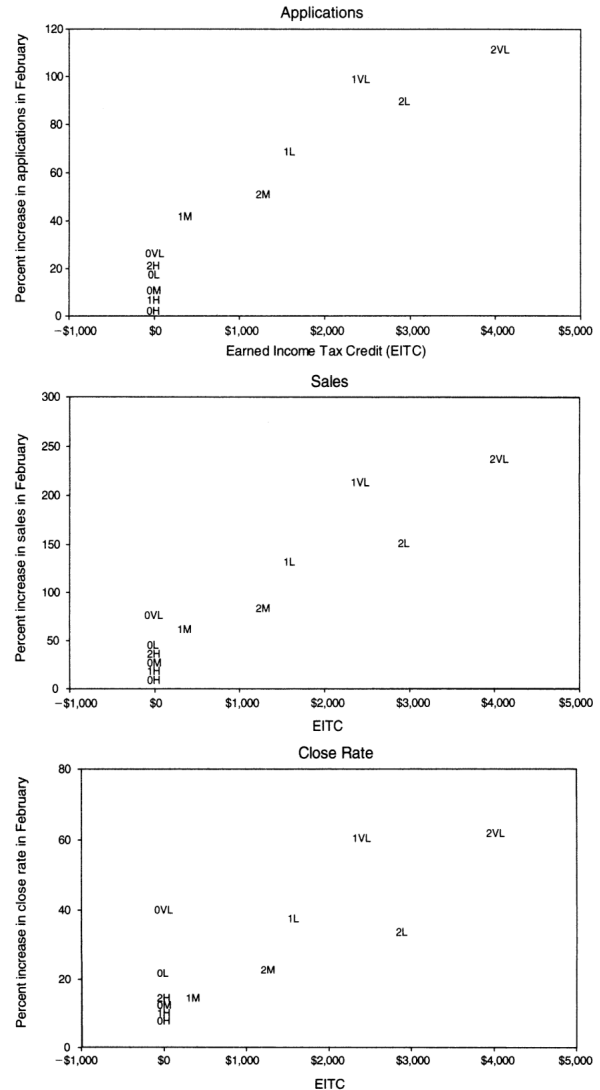


FIGURE 2B. TAX CREDIT EFFECTS ON APPLICATIONS AND SALES

(b) Figure 2B: Tax credit effects on applications, sales, and close rates.

6.4 Table 2: Purchasing estimates

Read:

- Table 2 estimates purchase demand using both individual-level and cell-level specifications.
- The minimum down-payment coefficient is large and negative across specifications.
- A \$100 increase in minimum down payment lowers purchase probability by about 0.03 in the individual-level estimates.
- The price coefficient is much smaller; in the preferred IV specification, the same purchase effect would require a much larger increase in financed price.
- The February dummy is positive and large, consistent with the tax-season liquidity shock.

Economic message: Consumers respond much more to immediate cash requirements than to total financed price. This is the main demand-side evidence that borrowers are liquidity constrained rather than merely impatient.

TABLE 2—PURCHASING ESTIMATES

	Probit estimates of individual-level purchasing (Dep. var. = sale indicator)				Cell-level estimates (Dep. var. = log(sales))	
	dF/dx (1)	dF/dx (2) ^a	dF/dx (3) ^b	dF/dx (4) ^c	Coef. (5) ^d	Coef. (6) ^e
<i>Offer variables</i>						
Negotiated price (\$100s)	−0.0002 (0.0002)	−0.0010 (0.0011)	−0.0022 (0.0007)	−0.0032 (0.0006)	−0.0061 (0.0016)	−0.0102 (0.0063)
Minimum down (\$100s)	−0.0301 (0.0006)	−0.0299 (0.0006)	−0.0298 (0.0006)	−0.0303 (0.0006)	−0.0895 (0.0039)	−0.0889 (0.0039)
Maximum interest rate (APR)	−0.0010 (0.0004)	−0.0013 (0.0006)	−0.0016 (0.0005)	0.0003 (0.0006)	−0.0033 (0.0033)	−0.0049 (0.0041)
Term (months)	−0.0008 (0.0004)	−0.0002 (0.0008)	0.0006 (0.0006)	0.0025 (0.0006)	−0.0001 (0.0025)	0.0021 (0.0044)
<i>Car characteristics</i>						
Car cost (\$100s)	0.0005 (0.0002)	0.0014 (0.0012)	0.0025 (0.0007)	0.0034 (0.0006)	0.0071 (0.0018)	0.0113 (0.0067)
Premium (cost > \$7,500)	0.0040 (0.0032)	0.0038 (0.0035)	0.0036 (0.0034)	0.0006 (0.0035)	0.0990 (0.0376)	0.0966 (0.0379)
Car age (years)	0.0008 (0.0007)	0.0008 (0.0006)	0.0008 (0.0006)	−0.0001 (0.0006)	0.0086 (0.0067)	0.0088 (0.0066)
Odometer (10,000s)	−0.0008 (0.0004)	−0.0008 (0.0004)	−0.0008 (0.0004)	−0.0008 (0.0004)	−0.0139 (0.0042)	−0.0144 (0.0041)
Lot age (months)	−0.0019 (0.0007)	−0.0034 (0.0022)	−0.0055 (0.0015)	−0.0071 (0.0010)	−0.0232 (0.0068)	−0.0308 (0.0133)
<i>Individual characteristics</i>						
Income (\$1,000s/month)	0.0245 (0.0008)	0.0250 (0.0010)	0.0258 (0.0008)	0.0284 (0.0009)	0.0983 (0.0060)	0.1014 (0.0075)
Age	0.0084 (0.0003)	0.0084 (0.0003)	0.0085 (0.0003)	0.0082 (0.0003)	0.0732 (0.0053)	0.0747 (0.0056)
Age squared	−0.0001 (3.7E−06)	−0.0001 (3.7E−06)	−0.0001 (3.8E−06)	−0.0001 (3.8E−06)	−0.0008 (6.7E−05)	−0.0008 (7.2E−05)
Bank account	0.0271 (0.0014)	0.0270 (0.0014)	0.0269 (0.0014)	0.0281 (0.0014)	−0.0010 (0.0296)	−0.0019 (0.0296)
House owner	−0.0320 (0.0018)	−0.0321 (0.0018)	−0.0321 (0.0018)	−0.0408 (0.0016)	−0.0171 (0.0367)	−0.0192 (0.0367)
Lives with parents	0.0091 (0.0021)	0.0090 (0.0021)	0.0089 (0.0021)	0.0097 (0.0021)	0.0391 (0.0387)	0.0339 (0.0379)
<i>Credit category fixed effects^f</i>						
Representative low risk	0.0269 (0.0070)	0.0264 (0.0070)	0.0256 (0.0070)	0.0239 (0.0070)	0.1333 (0.0517)	0.1304 (0.0519)
Representative medium risk	0.0394 (0.0063)	0.0397 (0.0063)	0.0402 (0.0063)	0.0381 (0.0064)	0.2974 (0.0434)	0.2975 (0.0435)
Representative high risk	0.0043 (0.0050)	0.0045 (0.0050)	0.0048 (0.0050)	0.0038 (0.0051)	0.0489 (0.0494)	0.0524 (0.0493)
<i>Month fixed effects^g</i>						
February (tax season)	0.1603 (0.0044)	0.1594 (0.0047)	0.1581 (0.0045)	0.1592 (0.0044)	0.5900 (0.0190)	0.5862 (0.0199)
Other fixed effects	Year, month, city, credit category	Year, month, city, credit category	Year, month, city, credit category	Year, month, city, credit category	Year, month, city, credit category	Year, month, city, credit category
Instrument for price	—	List price	Cost bucket dummies	State dummies	—	List price

Notes: Standard errors in parentheses. Sample for individual-level purchasing estimates is all applications; sample size is $N \gg 50,000$ (see Table 1). Sample size for cell-level estimates is $\approx 0.03N$. All standard errors are bootstrap estimates based on 50 random samples.

^a Instruments are list price (equal to zero if not available) and indicator equal to one if list price is not available. List prices are available for approximately 80 percent of the observations.

^b Instruments are dummy variables corresponding to each cost bucket (see Figure 3 for illustration).

Table 2: Purchasing estimates.

6.5 Figures 3 and 4: Car choice robustness and theory

Read:

- Figure 3 compares car-cost distributions before and after a pricing-policy change. The distributions barely move, suggesting little substitution toward cheaper cars after higher markups.
- Figure 4 lays out the theory. Moral hazard can make profits nonmonotone in future loan liability, generating loan caps. Adverse selection can make high-risk borrowers choose larger loans unless risk-based pricing separates types.

Economic message: Figure 3 supports the empirical focus on the buy/not-buy decision and loan size rather than rich car substitution. Figure 4 provides the mechanism linking credit-market

frictions to liquidity constraints.

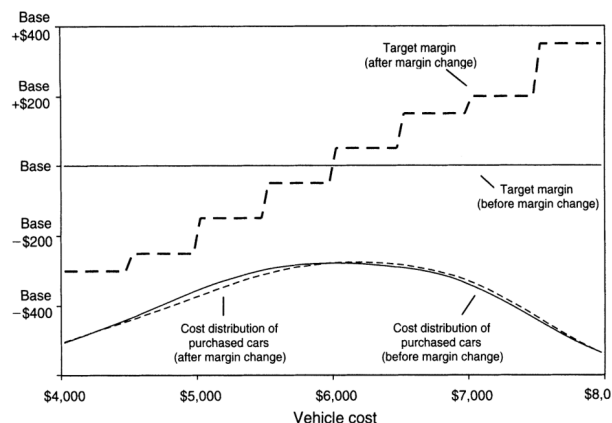


FIGURE 3. THE EFFECT (OR LACK THEREOF) OF PRICES ON CAR CHOICES

(a) Figure 3: The effect, or lack thereof, of prices on car choices.

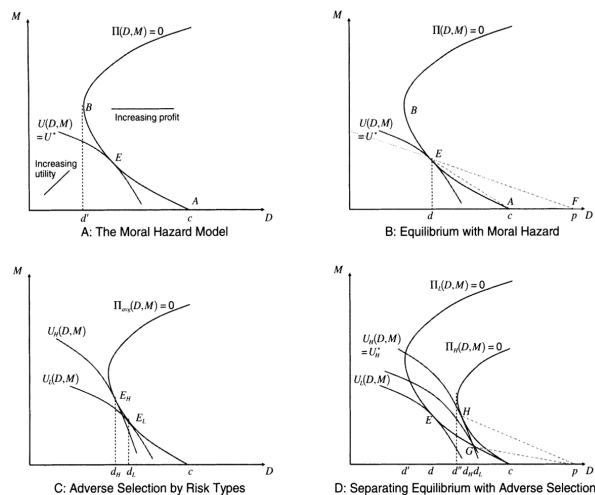


FIGURE 4. THEORY

(b) Figure 4: Theory.

6.6 Figures 5A and 5B: Repayment probabilities

Read:

- Figure 5A shows repayment probability falling with loan amount within each risk group.
- Low-risk borrowers have higher repayment probabilities than medium- or high-risk borrowers at comparable loan sizes.
- Figure 5B shows that borrowers making exactly the minimum down payment repay less often than borrowers paying more than the minimum.

Economic message: These descriptive patterns preview the paper's two central mechanisms: larger loans create repayment problems, and borrowers who choose minimum down payments are worse risks.

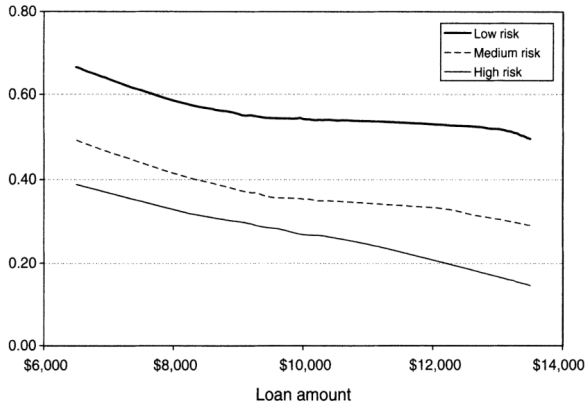


FIGURE 5A. PROBABILITY OF PAYMENT VERSUS LOAN AMOUNT

(a) Figure 5A: Probability of payment versus loan amount.

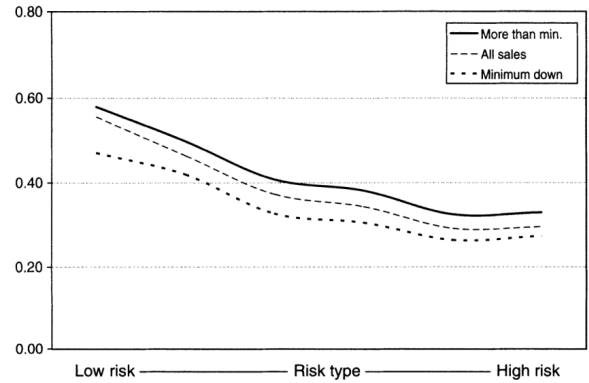


FIGURE 5B. PROBABILITY OF PAYMENT BY RISK TYPE AND DOWN PAYMENT

(b) Figure 5B: Probability of payment by risk type and down payment.

6.7 Tables 3 and 4: Down-payment demand and default hazard

Read:

- Table 3 estimates desired down payments among buyers. Higher-risk buyers choose smaller desired down payments, implying larger desired loans.
- February buyers choose higher desired down payments, consistent with increased cash on hand during tax season.
- Table 4 estimates default hazards. In the richest specification, a \$100 increase in amount financed raises the hazard by a factor of about 1.016; a \$1,000 increase raises the hazard by roughly 16 percent.
- The down-payment residual is below one in hazard-rate form, meaning borrowers who choose higher down payments for unobserved reasons are less likely to default.

Economic message: The tables quantify both adverse selection and moral hazard. Riskier borrowers want larger loans, and larger loans causally increase default.

TABLE 3—TOBIT ESTIMATES OF DOWN PAYMENT
(Dependent variable: Down payment (\$100s) conditional on purchase)

	Coefficient	Standard error	Coefficient	Standard error
	(1)		(2) ^a	
<i>Offer variables</i>				
Negotiated price (\$100s)	0.049	(0.004)	0.177	(0.002)
Maximum interest rate (APR)	0.203	(0.011)	0.242	(0.013)
Term (months)	-0.413	(0.009)	-0.503	(0.018)
<i>Car characteristics</i>				
Car cost (\$100s)	0.228	(0.004)	0.100	(0.002)
Premium (cost > \$7,500)	3.769	(0.078)	3.759	(0.079)
Car age (years)	0.061	(0.016)	0.065	(0.016)
Odometer (10,000s)	-0.026	(0.012)	-0.022	(0.012)
Lot age (months)	-0.559	(0.016)	-0.331	(0.044)
<i>Individual characteristics</i>				
Income (\$1,000s/month)	-0.164	(0.019)	-0.260	(0.026)
Age	-0.169	(0.001)	-0.186	(0.010)
Age squared	0.002	(1E-04)	0.002	(1E-04)
Bank account	0.202	(0.046)	0.235	(0.047)
House owner	0.226	(0.055)	0.241	(0.055)
Lives with parents	0.264	(0.055)	0.266	(0.056)
<i>Credit category fixed effects</i>				
Representative low risk	4.734	(0.119)	4.961	(0.126)
Representative medium risk	3.215	(0.107)	3.270	(0.108)
Representative high risk	0.718	(0.119)	0.733	(0.120)
<i>Month fixed effects</i>				
February (tax season)	3.171	(0.098)	3.259	(0.162)
Other fixed effects	Year, month, city, credit category		Year, month, city, credit category	
Instrument for price ^a	—		List price	

Notes: Sample is all sales; sample size is $\approx 0.34N$, where $N \gg 50,000$ (see Table 1). All results are based on Tobit regressions with actual down payment minus minimum down payment as the dependent variable and left-censoring at zero. Omitted fixed effects are “very high risk” for credit categories and “December” for months.

^a The results in column 2 are based on the joint maximum likelihood estimation of a negotiated price equation with list price as an explanatory variable and a down payment equation, assuming joint normality of the errors.

TABLE 4—PROPORTIONAL HAZARD MODEL ESTIMATES OF DEFAULT
(Dependent variable: Fraction of loan payments made)

	Hazard rate	Standard error	Hazard rate	Standard error	Hazard rate	Standard error	Hazard rate	Standard error
	(1)		(2)		(3)		(4)	
<i>Transaction characteristics</i>								
Amount financed (\$100s)	1.016	(0.001)	1.024	(0.001)	1.023	(0.001)	1.019	(0.000)
Maximum interest rate (APR)	1.022	(0.002)	1.026	(0.002)	1.025	(0.002)	1.022	(0.002)
Term (months)	1.015	(0.002)	1.006	(0.002)	1.008	(0.002)	1.008	(0.002)
Down payment residual (\$100s)	0.982	(0.001)	—	—	—	—	—	—
<i>Car characteristics</i>								
Car cost (\$100s)	0.981	(0.001)	0.975	(0.001)	0.974	(0.001)	0.976	(0.001)
Premium (cost > \$7,500)	0.867	(0.015)	0.888	(0.015)	0.887	(0.015)	0.819	(0.014)
Car age (years)	1.028	(0.003)	1.028	(0.003)	1.027	(0.003)	1.021	(0.003)
Odometer (10,000s)	1.012	(0.002)	1.012	(0.002)	1.012	(0.002)	1.015	(0.002)
Lot age (months)	1.055	(0.003)	1.065	(0.003)	1.067	(0.003)	1.062	(0.003)
<i>Individual characteristics</i>								
Income (\$1,000s/month)	0.955	(0.004)	0.948	(0.004)	—	—	—	—
Age	0.996	(0.002)	0.993	(0.002)	—	—	—	—
Age squared	1.000	(2E-05)	1.000	(7E-03)	—	—	—	—
Bank account	0.818	(0.007)	0.823	(0.007)	—	—	—	—
House owner	0.998	(0.011)	1.004	(0.011)	—	—	—	—
Lives with parents	1.059	(0.011)	1.060	(0.011)	—	—	—	—
<i>Credit category fixed effects</i>								
Representative low risk	0.518	(0.011)	0.509	(0.011)	0.461	(0.009)	—	—
Representative medium risk	0.801	(0.013)	0.789	(0.013)	0.748	(0.012)	—	—
Representative high risk	0.994	(0.018)	0.990	(0.018)	0.963	(0.017)	—	—
<i>Month fixed effects</i>								
February (tax season)	1.071	(0.022)	1.090	(0.022)	0.973	(0.023)	—	—
Other fixed effects	Year, month, city, credit category		Year, month, city, credit category		Year, month, city, credit category		Year, month, city, credit category	

Notes: Sample is all sales; sample size is $\approx 0.34N$, where $N \gg 50,000$ (see Table 1). The down payment residual in column 1 is computed from the Tobit regression presented in column 2 of Table 3. Omitted fixed effects are “very high

(a) Table 3: Tobit estimates of down payment.

(b) Table 4: Proportional hazard estimates of default.

6.8 Figures 6A and 6B: Expected payments and offer curves

Read:

- Figure 6A uses the default model to show expected loan payments as a function of loan amount. Expected payments rise initially but eventually fall as default risk increases.
- The peak occurs earlier for riskier borrowers.
- Figure 6B plots offer curves showing how future loan liability changes with down payment for different risk groups and prices.
- Higher-risk borrowers face higher required down payments, and borrowers who pay more down can receive lower effective interest rates.

Economic message: The lender’s contract menu is a rational response to default risk. Loan caps and minimum down payments are not arbitrary; they prevent marginal loans that are expected to lose money.

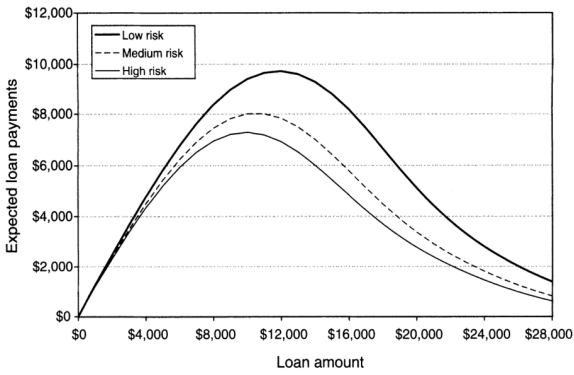


FIGURE 6A. EXPECTED LOAN PAYMENTS VERSUS LOAN AMOUNT

(a) Figure 6A: Expected loan payments versus loan amount.

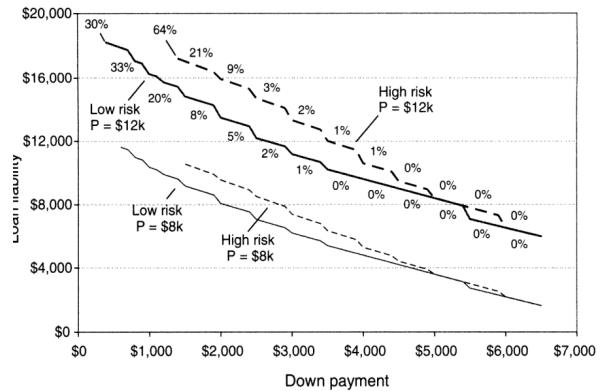


FIGURE 6B. OFFER CURVES

(b) Figure 6B: Offer curves.

6.9 Figures 7A and 7B: Desired versus actual down payments

Read:

- Figure 7A shows that desired down payments decline as default probability rises across credit categories.
- This is the adverse-selection force: high-risk borrowers would like to put less money down and borrow more.
- Figure 7B shows actual down payments after the lender's minimum requirements bind.
- The highest-risk categories are pushed toward larger actual down payments than they would choose voluntarily.

Economic message: Risk-based pricing mitigates adverse selection. Credit scoring does not merely predict default; it changes contract terms in a way that forces risky borrowers to hold smaller loans.

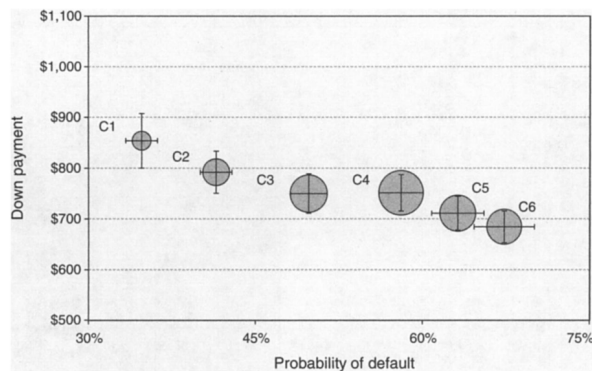


FIGURE 7A. SCATTER PLOT OF DESIRED DOWN PAYMENT VERSUS DEFAULT PROBABILITY BY CREDIT CATEGORY

(a) Figure 7A: Desired down payment versus default probability.

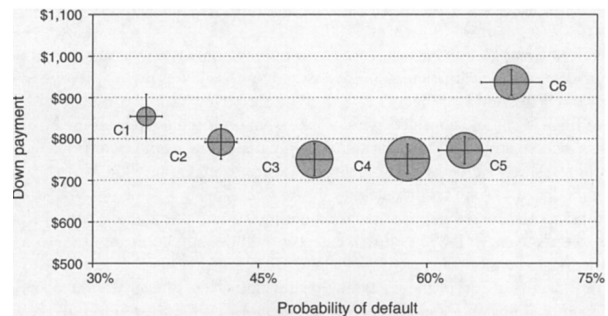


FIGURE 7B. SCATTER PLOT OF ACTUAL DOWN PAYMENT VERSUS DEFAULT PROBABILITY BY CREDIT CATEGORY

(b) Figure 7B: Actual down payment versus default probability.

7 Short summary

- The paper studies subprime auto lending using detailed application, transaction, and repayment data from a large lender.
- Buyers are low-income and often have poor credit histories; only about one-third of applicants purchase.
- Demand spikes during tax rebate season, especially for households with larger expected EITC refunds.
- Purchase decisions are far more sensitive to minimum down payments than to car prices.
- Default is very common and often occurs early.
- A \$1,000 larger loan raises the default hazard by about 16 percent for a given borrower.
- Higher-risk borrowers prefer smaller down payments and larger loans, creating adverse selection.
- Risk-based pricing forces riskier borrowers to make larger down payments, mitigating adverse selection across observable risk groups.
- Within risk groups, adverse selection remains because borrowers have private information about their likelihood of default.
- The paper’s central contribution is to connect household liquidity constraints to lender-side moral hazard, adverse selection, and contract design.

8 Conclusion

8.1 Core conclusion

Adams, Einav, and Levin show that liquidity constraints in subprime consumer lending are economically large and tightly connected to lender-side information problems. The demand evidence shows that borrowers are highly sensitive to cash on hand. The repayment evidence shows why lenders restrict credit: larger loans increase default, and riskier borrowers tend to demand larger loans.

8.2 Economic intuition

The intuition is simple. Many consumers want to buy cars but cannot easily produce cash for the down payment. They are therefore extremely sensitive to the immediate cash requirement. From the lender’s perspective, however, reducing the down payment is dangerous because it increases loan size. Larger loans make default more likely, and the people most eager to borrow more are often the riskiest borrowers. The lender responds by setting minimum down payments and using credit scoring to require more cash from high-risk borrowers.

8.3 Incremental contribution revisited

The paper advances the literature in three ways. First, it documents liquidity-constrained behavior in a setting with direct application and transaction data. Second, it connects that behavior to a structural credit-market mechanism rather than treating it as a reduced-form consumption anomaly.

Third, it separates moral hazard from adverse selection using exogenous variation in loan size and a control-function approach based on the down-payment decision.

8.4 Implications for consumer credit

The results imply that expanding subprime credit is difficult because lender constraints are not merely institutional conservatism. If borrowers default more when loan sizes rise, and if risky borrowers seek larger loans, then unconstrained credit can destroy lender value. Credit scoring helps, but mainly against observable selection. Moral hazard and unobservable selection remain.

8.5 Policy implications

Policies that reduce immediate liquidity needs, such as savings support or better-timed income transfers, may affect purchases strongly. Policies that simply encourage larger loans can raise default risk. Interest-rate caps may also matter because they limit the lender's ability to price risk, pushing adjustment into nonprice contract terms such as down-payment requirements and loan caps.

8.6 Limitations

The evidence comes from one company in the subprime used-car market. The paper does not fully observe outside borrowing options, household assets, or borrower expectations. It also does not separately identify whether default reflects strategic nonpayment, bad shocks, overoptimism, or other behavioral mechanisms. These limitations do not undermine the main market-equilibrium result, but they matter for welfare and policy interpretation.

8.7 Takeaway

The enduring takeaway is that subprime borrowers look liquidity constrained because they are very sensitive to cash up front, but lenders restrict credit for reasons that are economically meaningful. Liquidity constraints, moral hazard, adverse selection, and risk-based pricing are all part of the same market equilibrium.